

Madhav Umesh Kamble

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EDUCATION

Indian Institute of Information Technology Allahabad

M.Tech in Information Technology (Data Engineering)

- CGPA: 7.89 / 10
- Research: *Missing Data Imputation via Genetic Algorithm*
- Research: *Enhancing Genetic Algorithm with Explainable AI for Last-Mile Routing*

Prayagraj, Uttar Pradesh

Aug. 2025 – July 2027

SKN Sinhgad College of Engineering

B.Tech in Computer Science and Engineering

- CGPA: 9.31 / 10

Korti, Pandharpur

Aug. 2021 – May 2025

EXPERIENCE

Leadsoft IT Solutions

Full Stack Web Development Intern

Virtual

Aug. 2023 – Oct. 2023

- Built a Gate Pass Management mobile app (React Native + MERN stack) supporting digital pass requests, multi-level approvals, and real-time tracking.
- Developed RESTful APIs in Node.js/Express with MongoDB for pass lifecycle management and real-time status updates.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Java, C, C++, SQL, HTML, CSS

Data & ML: NumPy, Pandas, Scikit-learn, XGBoost, LightGBM, CatBoost, Matplotlib, Seaborn, Hadoop

Data Engineering: Apache Kafka, PySpark, Delta Lake, Apache Airflow, MLflow, Redis

Web Frameworks: React.js, Next.js, Node.js, Express.js, Tailwind CSS

Databases: MongoDB, MySQL, PostgreSQL

Tools & Platforms: Git, GitHub, Docker, Linux, VS Code

Relevant Coursework: Big Data Analytics, Data Visualisation, Machine Learning, Deep Learning, OS, CN, DBMS

PROJECTS

Real-Time Ride-Sharing Analytics Platform — [GitHub](#)

Jan. 2026 – May 2026

Python, Kafka, PySpark, Delta Lake, Airflow, XGBoost, MLflow, Redis, Streamlit, Docker

- Built an end-to-end streaming platform simulating Uber/Ola internals: Kafka ingestion into PySpark Structured Streaming, Delta Lake medallion architecture, ML-powered surge pricing, and a live Streamlit dashboard.
- Designed Bronze-Silver-Gold pipeline processing 8 events/sec across 3 Kafka topics; served XGBoost surge model via MLflow registry and PySpark Pandas UDF.
- Reduced dashboard latency from approx. 10s to under 1ms with Redis cache; orchestrated 4 Airflow DAGs (hourly refresh, daily retrain, quality checks, cache warmup) across 10 Docker services.

Traffic Demand Prediction — [GitHub](#)

Apr. 2026 – May 2026

Python, LightGBM, XGBoost, CatBoost, Pandas, NumPy

- Built an ML pipeline to predict normalized traffic demand across 1,200+ geographic zones, achieving R2 score of 91.8% – ranked Top 1.5% in a competitive hackathon.
- Engineered a 3-model GBT ensemble with pseudo-labeling, residual modeling, and geohash-based spatial features to overcome train/test distribution shift.

MIGA: Missing Data Imputation via Genetic Algorithm — [GitHub](#)

Jan. 2026 – Present

Python, NumPy, SciPy, scikit-learn, Pandas, Matplotlib, Genetic Algorithms

- Reimplemented MIGA (Figueroa-García et al., *Information Sciences* 2023) as the first open-source Python implementation; benchmarked against MICE, KNN, and mean imputation across 7 UCI datasets with Wilcoxon significance tests.
- Extended with Ledoit-Wolf shrinkage covariance, adaptive mutation scheduling, MNAR evaluation, variance preservation analysis, and a kurtosis-augmented fitness function (Fr+).

ACHIEVEMENTS

LeetCode: Rating 1669

CodeChef: Rating 1441

Runner-Up – Sinhgad Hackathon 2023

Runner-Up – Sinhgad Hackathon 2022

CERTIFICATIONS

0 to 100 Full-Stack Cohort – 100xDevs (2024)

Cyber Threat Intelligence 101 – arcX (2024)

Getting Started with Enterprise Data Science – IBM (2024)